PROJECT REPORT · COMPILED OCTOBER 25, 2026

AI in Healthcare Corporate Finance

How health systems use AI in forecasting, accounting, audit and the revenue cycle, and what a five-year business case looks like for a $2 billion health system.

Summary

The base case returns $51.2M in benefits on $11.8M of cost over five years, with payback in year 2. Even the conservative case stays positive.

CONSERVATIVE NPV	BASE NPV	OPTIMISTIC NPV
$3.8M	$29.2M	$71.1M
Payback year 4 · return on cost 50%	Payback year 2 · return on cost 336%	Payback year 1 · return on cost 914%

Adoption is broad, measurement is not.

77% of finance organizations use AI, yet only 35% say they measure its ROI effectively (Protiviti, 2026).

The revenue cycle has the strongest evidence.

At least 60% of health systems used AI or related technology to manage denials in 2025, up from 23% in 2024 (McKinsey, via Modern Healthcare). In the base case, denials and receivables supply 60% of benefits.

FP&A and accounting benefits rest on assumptions.

Forecasting is the top AI use case in finance, but the sources reviewed give no quantified healthcare outcomes for it. The model sets those drivers low on purpose.

Data foundations are the constraint.

The 2026 FP&A Trends Survey (n=475) finds AI is moving into finance faster than the foundations to support it, and 61% of finance leaders still use spreadsheets as their main forecasting tool.

Where AI is used today

Three areas of healthcare corporate finance, each with a different level of evidence.

FP&A AND FORECASTING

Forecasting is the leading use case

AI forecasting adoption rose from 58% to 76% in a year across finance organizations (Protiviti). Improving forecast accuracy is the top FP&A priority for 35% of finance leaders (Vena). In healthcare, finance teams describe moving from spreadsheet-driven, periodic forecasts toward predictive forecasting and AI-assisted analysis.

- 21% of FP&A teams use generative AI and another 25% plan to; 12% use machine learning in forecasting (FP&A Trends Group).
- Evidence type: surveys for adoption, no quantified healthcare outcomes found.

ACCOUNTING, AUDIT AND COMPLIANCE

Anomaly detection and continuous auditing

AI is applied to billing-pattern anomalies, duplicate and phantom claims, and audit preparation. KPMG's view is that analytics and AI can strengthen fraud detection, risk assessment and continuous auditing for internal audit teams.

- Most quantified claims come from vendors, such as a 96% fraud-detection accuracy figure, and the model excludes them.
- Compliance sign-off and sampling audits remain necessary because AI-coded claims must still meet CMS, HIPAA and payer rules.

REVENUE CYCLE

Denial prevention, coding and receivables

Health systems use AI to predict and prevent denials, assist coding and automate appeals. 80% were exploring, piloting or implementing generative AI in revenue cycle in 2025, with 27% at scale (HFMA/AKASA, as summarized by Stealth Agents).

- One nonprofit system cut days in receivables from 53 to 47 between 2023 and 2025 (Modern Healthcare).
- Reported denial reductions include 25% at one hospital and 42% for registration and eligibility denials at OhioHealth. Both are vendor-reported.

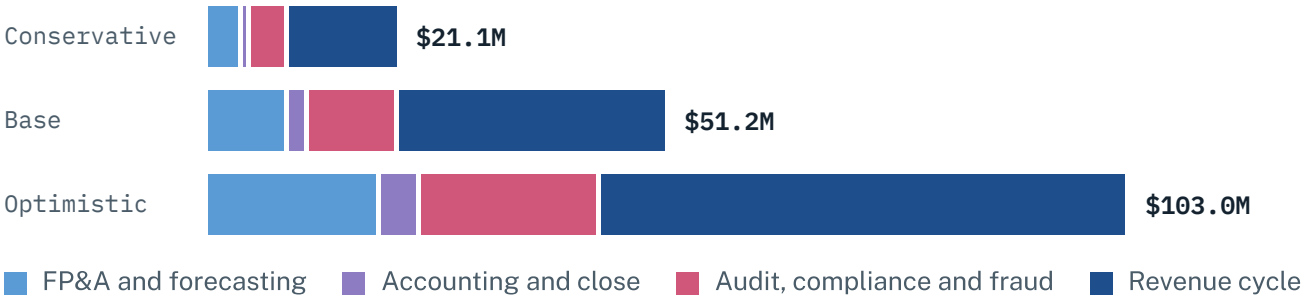
Five-year ROI model

An illustrative health system with $2.0B in net patient revenue. Every driver is editable in the accompanying Excel workbook, and the three scenarios share the same logic.

RESULT	CONSERVATIVE	BASE	OPTIMISTIC
Total benefits	$21.1M	$51.2M	$103.0M
Total costs	$14.1M	$11.8M	$10.2M
Net present value (8% discount rate)	**$3.8M**	**$29.2M**	**$71.1M**
Return on cost (net benefit / total cost)	50%	336%	914%
Payback	Year 4	Year 2	Year 1
Year 5 net benefit, annual	$4.7M	$12.0M	$23.7M

Where the benefits come from

Five-year benefits by use case, $ millions

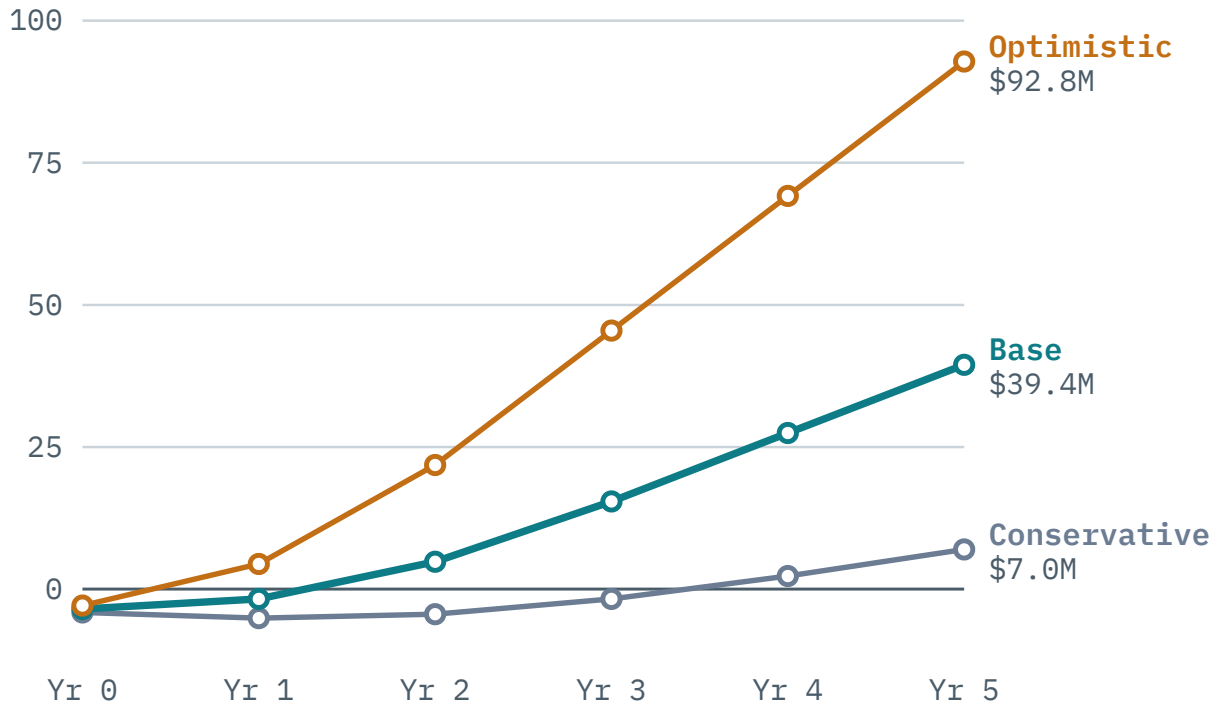


USE CASE	CONSERVATIVE	BASE	OPTIMISTIC
FP&A and forecasting	$3.6M	$8.8M	$19.0M
Accounting and close	$0.5M	$1.9M	$4.1M
Audit, compliance and fraud	$4.1M	$9.8M	$20.1M
Revenue cycle	$12.9M	$30.7M	$59.7M
Total	**$21.1M**	**$51.2M**	**$103.0M**

Denial write-off reduction alone is 54% of base-case benefits, so the result is most sensitive to the denial rate, the write-off share and the reduction achieved.

When the investment pays back

Cumulative net benefit by year, $ millions, before discounting



Base case by year

$ MILLIONS	YEAR 0	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	5-YR TOTAL
BENEFITS							
FP&A time savings	-	$0.1M	$0.3M	$0.4M	$0.5M	$0.5M	$1.7M
Forecast accuracy	-	$0.5M	$1.1M	$1.7M	$1.9M	$1.9M	$7.1M
Accounting and close time savings	-	$0.1M	$0.3M	$0.5M	$0.5M	$0.5M	$1.9M
Audit and compliance efficiency	-	$0.1M	$0.4M	$0.5M	$0.6M	$0.6M	$2.2M
Billing-error and compliance exposure	-	$0.5M	$1.2M	$1.8M	$2.0M	$2.0M	$7.5M
Denial write-off reduction	-	$1.8M	$4.4M	$6.6M	$7.3M	$7.3M	$27.4M
Accounts receivable carrying benefit	-	$0.2M	$0.5M	$0.8M	$0.9M	$0.9M	$3.3M
Total benefits	**-**	**$3.4M**	**$8.2M**	**$12.3M**	**$13.7M**	**$13.7M**	**$51.2M**
COSTS							
Implementation and training	$3.5M	-	-	-	-	-	$3.5M
Software and licensing	-	$1.2M	$1.2M	$1.2M	$1.2M	$1.2M	$6.0M
Governance staff	-	$0.5M	$0.5M	$0.5M	$0.5M	$0.5M	$2.2M
Total costs	**$3.5M**	**$1.6M**	**$1.6M**	**$1.6M**	**$1.6M**	**$1.6M**	**$11.8M**
Net benefit	**($3.5M)**	**$1.8M**	**$6.5M**	**$10.6M**	**$12.0M**	**$12.0M**	**$39.4M**
Cumulative net benefit	($3.5M)	($1.7M)	$4.8M	$15.4M	$27.4M	$39.4M	

Assumptions

The first block describes the illustrative organization. Replace it with real figures to tailor the model. The remaining rows differ by scenario.

DRIVER	CONSERVATIVE	BASE	OPTIMISTIC
A. ORGANIZATION PROFILE (ILLUSTRATIVE MID-SIZE HEALTH SYSTEM - REPLACE WITH YOUR OWN FIGURES)			
Net patient service revenue		$2,000.0M	
Operating expenses as % of net revenue		95%	

DRIVER	CONSERVATIVE	BASE	OPTIMISTIC
Initial claim denial rate (% of billed revenue)		12.2%	
Share of denied dollars ultimately written off		15%	
FP&A / reporting FTEs in scope		25	
Accounting & close FTEs in scope		35	
Fully loaded cost per finance FTE		$120,000	
Annual external audit, internal audit & compliance program cost		$6.0M	
Annual billing-error & compliance exposure (% of net revenue)		1%	
Cost of capital (used for AR carrying benefit and NPV discounting)		8%	
B. AI BENEFIT DRIVERS (SCENARIO-SPECIFIC)			
FP&A: share of team time freed by AI	15%	25%	35%
Accounting & close: share of team time freed by AI	10%	20%	30%
Realization of freed time as cash savings	40%	60%	80%
Forecast accuracy: avoidable cost variance eliminated (% of operating expenses)	0.05%	0.10%	0.20%
Audit & compliance cost efficiency	5%	10%	15%
Reduction in billing-error / compliance exposure	5%	10%	20%
Reduction in denial write-offs	10%	20%	35%
Reduction in days in accounts receivable	1.0	2.0	4.0
C. COSTS (SCENARIO-SPECIFIC)			
One-time implementation & integration	$3.5M	$3.0M	$2.5M
One-time training & change management	$0.6M	$0.5M	$0.4M
Annual software / cloud / model licensing	$1.4M	$1.2M	$1.0M
Governance & model-monitoring FTEs	4	3	3
Cost per governance FTE		$150,000	
D. BENEFIT RAMP-UP (SHARE OF FULL RUN-RATE BENEFIT REALIZED IN EACH YEAR)			
Year 1	15%	25%	35%
Year 2	40%	60%	75%
Year 3	70%	90%	100%
Year 4	90%	100%	100%

DRIVER	CONSERVATIVE	BASE	OPTIMISTIC
Year 5		100%	

Read this before relying on the numbers. Organization size, headcount, cost per FTE, the write-off share, exposure and all time-saving and cost drivers are planning assumptions I chose, not sourced figures. Only the denial rate, the denial-reduction range and the receivables-days range are anchored to published results, and those come from vendors and single cases.

Evidence base

Benchmarks collected for the project. Filled tags mark surveys and press reports of surveys. Outlined tags mark vendor-reported or secondary figures, which should be treated as upper bounds.

AREA	FINDING	VALUE	SOURCE	EVIDENCE
General finance	Finance organizations employing AI	**77%**	Protiviti 2026 Global Finance Trends Survey	Survey
FP&A	AI adoption for financial forecasting, year over year	**58% to 76%**	Protiviti 2026 Global Finance Trends Survey	Survey
General finance	Finance organizations effective at measuring AI ROI	**35%**	Protiviti 2026 Global Finance Trends Survey	Survey
FP&A	FP&A teams using generative AI (further 25% planning to adopt)	**21%**	FP&A Trends Group 2026 survey, sponsored by OneStream (n=475)	Survey
FP&A	FP&A teams using machine learning in forecasting	**12%**	FP&A Trends Group 2026 survey (OneStream)	Survey
FP&A	Finance leaders saying the C-suite has mandated AI use in finance	**70%**	Vena 7 FP&A Trends Shaping 2026 (n>400)	Vendor survey
FP&A	Finance leaders with AI agents fully integrated across FP&A	**34%**	Vena 7 FP&A Trends Shaping 2026	Vendor survey
FP&A	Finance leaders using spreadsheets as primary budgeting/forecasting tool	**61%**	Vena 7 FP&A Trends Shaping 2026	Vendor survey

AREA	FINDING	VALUE	SOURCE	EVIDENCE
FP&A	Improving forecast accuracy as top FP&A priority (72% place in top three)	**35%**	Vena 7 FP&A Trends Shaping 2026	Vendor survey
Revenue cycle	Health systems exploring, piloting or implementing GenAI in RCM (27% at scale, 53% piloting)	**80%**	HFMA/AKASA 2025 survey, as summarized by Stealth Agents	Secondary
Revenue cycle	Health systems using AI/technology for denials (2025 vs 2024: 23%) and prior authorization (2024: 44%)	**At least 60%**	McKinsey, via Modern Healthcare (Jan 30 2026)	Press report of survey
Revenue cycle	Revenue cycle leaders using advanced technology to improve coding (2024: 39%)	**More than 50%**	McKinsey, via Modern Healthcare (Jan 30 2026)	Press report of survey
Revenue cycle	Average US health-system claim denial rate, 2025	**12.2%**	Delos Intelligence blog (Aug 2026)	Vendor-reported
Revenue cycle	Days in AR at one nonprofit system with AI-driven applications (2023 to 2025)	**53 to 47 days**	Modern Healthcare, via CHA daily news clip (Jan 30 2026)	Press report, single case
Revenue cycle	OhioHealth reduction in registration/eligibility-related denials	**42%**	EngineerBabu citing Experian Health	Vendor-reported / secondary
Revenue cycle	Providers using AI in claims processes who report reduced denials	**69%**	Experian Health 2025 survey, via EngineerBabu	Secondary
Revenue cycle	Reported reduction in preventable denials by AI prevention tools	**30% to 70%**	EngineerBabu	Vendor-reported

AREA	FINDING	VALUE	SOURCE	EVIDENCE
Revenue cycle	Claim denials reduction at one hospital using Jorie AI tools in six months	**25%**	Simbo AI blog	Vendor case
Revenue cycle	Annual revenue lost to revenue cycle inefficiency (US)	**$262 billion**	Stealth Agents / Delos / EngineerBabu	Vendor-reported
Audit & compliance	AI and data analytics can help internal audit enhance fraud detection, risk assessment and continuous auditing	**Qualitative**	KPMG, via Optro blog	Consultancy view
Audit & compliance	Average healthcare fraud scheme runs 18-24 months before detection via traditional audit	**18-24 months**	Oxmaint blog	Vendor-reported

Risks and governance

What could reduce the benefits or create new exposure, with suggested mitigations and owners.

Data and process readiness

The 2026 FP&A Trends Survey concludes AI is outpacing the foundations needed to use it; 61% of finance leaders still rely on spreadsheets as the primary forecasting tool (Vena).

Mitigation. Fix data definitions and integrations first; start with well-instrumented processes; track data-quality KPIs.

Owner: CFO / Finance systems

Unmeasured ROI

Only 35% of finance organizations say they measure AI ROI effectively (Protiviti).

Mitigation. Set baselines before launch; assign a benefit owner per use case; review realized vs modelled benefit quarterly.

Owner: FP&A lead

Model errors in financial outputs

Generative and predictive models can produce wrong forecasts, narratives or coding suggestions that flow into reporting or claims.

Mitigation. Human review of material outputs, reconciliation to source, model validation and back-testing, clear escalation paths.

`Owner: Controller / Internal audit`

Regulatory and false-claims exposure

AI-assisted coding and billing must still meet CMS, HIPAA and payer rules; automated errors can scale quickly.

Mitigation. Compliance sign-off, sampling audits of AI-coded claims, documented audit trail, vendor attestations.

`Owner: Chief Compliance Officer`

Payer response (AI arms race)

Payers also use AI for review; documentation that satisfied criteria in 2022 can fail AI-driven review in 2026 (EngineerBabu).

Mitigation. Invest in documentation quality and payer-rule monitoring; avoid assuming denial gains are permanent.

`Owner: VP Revenue Cycle`

Privacy and security

Financial and patient-linked data feed these models; third-party AI vendors expand the attack surface.

Mitigation. Vendor security review, minimum-necessary data, access controls, contractual limits on data use.

`Owner: CISO / Privacy officer`

Integration and vendor lock-in

Many systems prefer solutions embedded in the EHR or ERP; point tools can fragment workflows.

Mitigation. Define architecture standards, data-portability terms and exit clauses before selecting vendors.

`Owner: CIO / Finance systems`

Workforce and change management

Roles shift from data production to AI oversight; realization of time savings depends on adoption.

Mitigation. Training, redeployment plans, clear communication; model realization rate conservatively (see Assumptions).

`Owner: CFO / HR`

Recommendations

My reading of the evidence and the model results.

1. **Start with the revenue cycle.** It has the strongest published evidence and supplies most of the modelled value. Begin with denial prevention and receivables.
2. **Set baselines and a benefit owner before launch.** Only 35% of finance organizations measure AI ROI well, so record denial rate, days in receivables and close time first, then compare against the model each quarter.

3. **Fix data and process foundations in parallel.** Forecasting and close automation depend on clean, integrated data, and many teams still forecast in spreadsheets.
4. **Build governance into the cost case.** Budget for model validation, human review of material outputs and compliance sampling of AI-coded claims.
5. **Replace the illustrative inputs.** Put in your own revenue, denial rate, headcount and vendor quotes, and treat the conservative case as the planning case until pilots prove more.

Limitations

The organization is illustrative, not a real health system. Several statistics come from vendor blogs or secondary summaries and were not independently verified; definitions such as the denial rate differ across sources. Web sources were retrieved in October 2026 and will date. The model estimates direct cost and revenue effects only, and omits patient experience, clinician burden and strategic value. This report is for learning and discussion and is not financial or legal advice.

Sources

- Protiviti 2026 Global Finance Trends Survey (press release via PR Newswire)
- OneStream / FP&A Trends Group: The Readiness Gap - 2026 FP&A Trends Survey
- Vena: 7 FP&A Trends Shaping 2026
- Strata Decision: AI in healthcare FP&A
- Modern Healthcare via CHA: Health systems bet on AI to curb denials and protect cash (Jan 30 2026)
- Stealth Agents: AI revenue cycle management automation statistics 2026 (summarizing HFMA/AKASA 2025)
- Delos Intelligence: AI-powered revenue cycle management (Aug 2026)
- EngineerBabu: How to reduce claim denials with AI (cites Experian Health)
- HFMA: Predict, prevent, perform - the AI evolution of denials management (Apr 2026)
- Simbo AI: How AI ensures compliance and reduces audit risks in medical billing
- Optro: Strengthen internal audits in healthcare (cites KPMG)
- Oxmaint: AI for medical fraud detection

Compiled from the project research and the AI_Healthcare_Finance_ROI_Model.xlsx workbook. All model figures on this page are read from that workbook after recalculation.